

LAMP Stack Setup and AWS Ubuntu Checks

Only the commands from the original instructions, with a short explanation for each

Part 1: Set Up the LAMP Stack

1. Update Ubuntu

```
sudo apt update  
sudo apt upgrade
```

`sudo apt update` refreshes Ubuntu's package list. `sudo apt upgrade` installs the available package updates.

2. Install Apache

```
sudo apt install apache2
```

Installs the Apache web server package on Ubuntu.

```
sudo systemctl start apache2  
sudo systemctl enable apache2
```

The first command starts Apache now. The second makes Apache start automatically when Ubuntu boots.

```
sudo systemctl status apache2
```

Displays Apache's current status so you can confirm that the service is running.

```
http://localhost
```

Open this address in a browser on the same machine to view the Apache default page.

3. Install MySQL

```
sudo apt install mysql-server
```

Installs the MySQL database server and its required packages.

```
sudo systemctl status mysql
```

Shows whether the MySQL service is running.

```
sudo mysql_secure_installation
```

Starts MySQL's interactive security setup for removing unsafe defaults and applying basic protection.

4. Install PHP

```
sudo apt install php libapache2-mod-php php-mysql
```

Installs PHP, Apache's PHP module, and the PHP connector for MySQL.

```
sudo systemctl restart apache2
```

Restarts Apache so it loads the newly installed PHP module.

5. Test PHP

```
sudo nano /var/www/html/info.php
```

Opens a new PHP test file in Nano inside Apache's website folder.

```
<?php  
phpinfo();  
?>
```

Enter this exact PHP code in the file. It produces a page containing PHP configuration information.

Save with Ctrl + O, press Enter, and exit with Ctrl + X.

```
http://localhost/info.php
```

Open this address in the browser. A PHP information page confirms that PHP is working with Apache.

6. Create a Project Folder

```
sudo mkdir /var/www/html/myproject  
sudo chown -R $USER:$USER /var/www/html/myproject
```

The first command creates the project folder. The second gives your current user ownership of that folder.

```
nano /var/www/html/myproject/index.php
```

Opens the project's first PHP file in Nano.

Create the First PHP Page

```
<?php
echo "Hello, LAMP is working!";
?>
```

Enter this exact PHP code. It displays a simple success message when the page runs.

```
http://localhost/myproject
```

Open this address to run the project and see the message from index.php.

7. Important Security Step

```
sudo rm /var/www/html/info.php
```

Deletes the PHP information page after testing because it exposes server configuration details.

Part 2: Connect to AWS Ubuntu and Check LAMP

1. Connect to Ubuntu Console

```
ssh -i your-key.pem ubuntu@your-public-ip
```

Connects from your local terminal to the Ubuntu EC2 instance using your private key and public IP address.

Example

```
ssh -i lamp-key.pem ubuntu@13.201.55.20
```

This is the original example showing the key filename and public IP in their correct positions.

```
chmod 400 your-key.pem
```

Restricts access to the private key. Run it if SSH reports that the key file's permissions are too open.

2. Check You Are on Ubuntu

```
lsb_release -a
```

Displays the Linux distribution name and Ubuntu version details.

or

```
cat /etc/os-release
```

Displays operating-system identification details stored by Ubuntu.

3. Check Apache

```
sudo systemctl status apache2
```

Shows whether Apache is running on the AWS Ubuntu server.

```
sudo systemctl start apache2
```

Starts Apache if the status command shows that it is not running.

```
http://your-public-ip
```

Open this address in your local browser after replacing the placeholder with the EC2 public IP.

Example

```
http://13.201.55.20
```

This is the original example of opening the Apache website using an EC2 public IP.

4. Check AWS Security Group

```
SSH TCP 22 Your IP  
HTTP TCP 80 0.0.0.0/0  
HTTPS TCP 443 0.0.0.0/0
```

These are the original inbound rules: SSH from your IP, plus public HTTP and HTTPS access.

5. Check PHP

```
echo "<?php phpinfo(); ?>" | sudo tee /var/www/html/info.php
```

Creates the PHP information test file inside Apache's website folder.

```
http://your-public-ip/info.php
```

Open this address after replacing the placeholder. The PHP information page confirms that PHP is working.

```
sudo rm /var/www/html/info.php
```

Deletes the PHP information page after the test so it does not expose server details.

6. Check MySQL

```
sudo systemctl status mysql
```

Displays the MySQL service status on the EC2 instance.

```
sudo mysql
```

Opens the MySQL command prompt using Ubuntu's administrative access.

```
SHOW DATABASES;  
EXIT;
```

The first command lists available databases. The second closes MySQL and returns to the Ubuntu terminal.

Quick Health Check Commands

```
sudo systemctl status apache2  
sudo systemctl status mysql  
php -v  
mysql --version
```

These exact commands check Apache, check MySQL, print the PHP version, and print the MySQL client version.